🐾 BIOLOGY CH.17 — ANIMAL KINGDOM — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this chapter:

1. Amoeba — Feeding & Pseudopodia

- Amoeba apni 'entire body surface area' se food leta hai — pseudopodia food particle ke around push karke usse engulf karta hai. Paramecium cilia se food leta hai. *[Q1008]*
- Pseudopodia (finger-like extensions) 'Amoeba' mein hote hain — cell membrane ke temporary projections, movement aur food catching ke liye. Amoeba constantly shape change karta rehta hai. *[Q1016]*

2. Parasitic Nutrition — Without Killing the Host

- 'Louse, tapeworm, leech, dodder' — group hai jo doosre plants-animals se nutrition leta hai bina unhe maare — Parasites host se nutrients extract karte hain. Types: Ectoparasites, Endoparasites, Meso Parasites. *[Q1009]*
- Tapeworm life cycle mein 'Pig' host hota hai (pork tapeworm/Taenia solium ke liye) — tapeworms Platyhelminthes phylum mein hain, proglottids add karke grow karte hain. *[Q1026]*
- 'Tapeworm' endoparasite ka example hai — host ke andar tissues-organs mein rehta hai. Ectoparasites (ticks, lice, fleas) skin ki superficial layers ko infect karte hain. *[Q1027]*

3. Fish — External Structure & Fins

- Fish diagram mein Gills (Position 1) respiration ke liye hote hain — Dorsal Fins rolling se protect karte hain, Pectoral Fins directional movement, Pelvic Fin stabilize karta hai. *[Q1010]*

4. Malpighian Tubules — Arthropoda

- 'Malpighian tubules' Arthropoda ki characteristic hai — excretory structures hain (lobsters, crabs, spiders, insects, centipedes mein). Echinodermata mein Starfish, Annelida mein Earthworms hain. *[Q1011]*

5. Metamorphosis

- Silk worms aur frog larvae ka adult mein transformation 'Metamorphosis' kehlaata hai — drastic rapid physical changes birth ke kuch samay baad hoti hain. *[Q1012]*

6. Algae as Space Food

- 'Spirulina aur Chlorella' algae protein-rich hain, space travellers ke food supplements ki tarah use hote hain — Chlorella 30% proteins-30% carbs, Spirulina protein-Vitamin B ka richest source hai. *[Q1013]*

7. Invertebrates vs Vertebrates

- ⚠ Trap: 'Reptiles' invertebrates mein SHAAMIL NAHI hain — reptiles vertebrates hain (backbone hota hai). Invertebrates: molluscs, arachnids, insects (koi backbone nahi). *[Q1014]*

8. The 5-Kingdom Classification — Animalia/Monera/Protista

- 'Metazoa' Kingdom Animalia mein classify hote hain — sabhi multicellular animals (sponges chhodkar), heterotrophic hote hain. 5 Kingdoms: Monera, Protista, Fungi, Plantae, Animalia. *[Q1015]*

- ⚠️ Trap: 'Protozoa' Whittaker ke 5-Kingdom classification ka part NAHI hai — ye Protista kingdom ka sub-category hai. 5 Kingdoms: Monera, Protista, Fungi, Plantae, Animalia (R.H. Whittaker, 1969). *[Q1033]*
- Whittaker classification mein 'Monera' group mein fully developed nucleus NAHI hota — prokaryotic hain (bacteria, blue-green algae), naked DNA. Baaki 4 kingdoms eukaryotic hain. *[Q1034]*
- 'Monera' group unicellular prokaryotic organisms (bina central membrane ke) hain jaise bacteria — 70s ribosomes, naked DNA, peptidoglycan cell wall. *[Q1052]*
- Multicellular organisms bina cell wall ke 'Animalia' kingdom mein aate hain — Plantae cellulose wall, Fungi chitin wall, Monera peptidoglycan wall rakhte hain. *[Q1063]*
- ⚠️ Trap: 'Diatom' Monera se related NAHI hai — ye eukaryotic hai, Kingdom Protista mein aata hai. Monera mein sirf prokaryotes (bacteria, Cyanobacteria/Anabaena, Archaebacteria) hote hain. *[Q1073]*

9. Cartilaginous Fish

- 'Shark' cartilaginous fish (Chondrichthyes) hai — skeleton cartilage se bana hota hai (Skates, Rays, Chimaeras bhi). Bony fish (Osteichthyes) mein Tuna, Rohu, Salmon aate hain. *[Q1017]*

10. Saprophytic Fungi — Penicillium

- 'Penicillium' saprophytic fungus hai — dead-decaying organic matter se nutrition leta hai (molds, mushrooms, yeast, mucor bhi). Penicillin isi se derive hoti hai (antibacterial drug). *[Q1018]*

11. Taxonomic Hierarchy

- 'Genus' Family aur Species ke beech ka classification level hai — hierarchy: Domain > Kingdom > Phylum > Class > Order > Family > Genus > Species. Carolus Linnaeus 'Father of taxonomy' hain. *[Q1019]*
- Animal classification ki hierarchy: Kingdom > Phylum > Class > Order > Family > Genus > Species — Carolus Linnaeus ne establish ki, broad se specific tak. *[Q1037]*

12. Arthropoda — Largest Group with Jointed Legs

- 'Arthropod' phylum ke animals ke jointed legs hote hain — largest-diverse group hai, jointed appendages-exoskeleton (insects, spiders, centipedes, millipedes). *[Q1020, Q1061]*
- 'Arthropoda' animal kingdom ka largest group hai — over two-thirds sabhi named species arthropods hain, segmented bodies with hard exoskeletons-jointed limbs. *[Q1039, Q1050, Q1065, Q1079]*

13. Cnidaria — Corals

- Corals marine invertebrates hain, class Anthozoa mein, phylum 'Cnidaria' ka hissa — hydras, jellyfish, corals, sea anemones bhi is phylum mein aate hain. *[Q1021]*

14. Mammals — Heart & Characteristics

- ⚠ Trap: Mammals ka heart 'three chambered' NAHI hai — FOUR-chambered hota hai. Mammals hair rakhte hain, mammary glands se doodh dete hain, kuch (monotremes) ande dete hain. *[Q1022]*
- Mammals 'warm-blooded' hote hain 4-chambered heart ke saath — oxygenated aur deoxygenated blood poori tarah alag hoti hai, highly efficient circulatory system. *[Q1040, Q1051]*

15. Mollusca

- ⚠ Trap: 'Scolopendra' (centipede) Mollusca ka example NAHI hai — Arthropoda phylum (Chilopoda class) mein aata hai. Mollusca mein Union, Octopus, Snail aate hain (2nd largest phylum). *[Q1023]*
- ⚠ Trap: 'Antedon' (feather star) Mollusca ka part NAHI hai — Echinodermata phylum mein aata hai. Mollusca mein Snail, Octopus, Chiton hote hain. *[Q1070]*

16. Pisces — Characteristics

- ⚠ Trap: 'Pseudocoelom' Pisces ki characteristic NAHI hai — ye Aschelminthes (roundworms) ki feature hai. Pisces cold-blooded hain, scales-gills-bony/cartilaginous endoskeleton rakhte hain. *[Q1031]*
- ⚠ Trap: Pisces 'warm-blooded' NAHI hote — cold-blooded aquatic animals hain, gills se oxygen lete hain, scales se covered hote hain, heart 2 chambers ka hota hai. *[Q1024]*
- Fish mein '2-chambered heart' hota hai (1 atrium + 1 ventricle) — single circulation hoti hai, blood heart se ek hi baar guzarta hai poore circuit mein. *[Q1049]*

17. Regeneration — Earthworm

- 'Earthworm' cut hone par regenerate ho sakta hai — Annelida phylum, clitellum ke peeche cut hone par head portion survive-regenerate kar sakta hai (Axolotl, Starfish bhi regenerate karte hain). *[Q1025]*

18. Animal Breeds — Goats

- 'Angora, Alpine aur Nubian' goat breeds hain — French-Alpine, Boer, Saanen, LaMancha bhi goat breeds hain. Cow breeds: Holstein, Jersey. Buffalo breeds: Murrah, Nili-Ravi. *[Q1028]*

19. Warm-Blooded vs Cold-Blooded Animals

- 'Pigeon' (bird) warm-blooded hai — Birds (Aves) aur Mammals constant body temperature maintain kar sakte hain. Fish, toad (amphibian), crocodile (reptile) cold-blooded hain. *[Q1029]*
- 'Aves' (birds) group warm-blooded hoti hai — Mammals bhi warm-blooded. Reptiles, Fish, Amphibians cold-blooded hote hain. *[Q1038, Q1041]*

20. Ultrasound — Echolocation

- 'Porpoises' ultrasound produce kar sakte hain (>20 KHz) — Dolphins, Bats bhi echolocation ke liye ultrasound use karte hain. Elephants-Rhinoceroses infrasound (<20 Hz) produce karte hain. *[Q1030]*

21. Annelida

- ⚠ Trap: 'Ascaris' Annelida class ka part NAHI hai — Phylum Nematoda (roundworm) mein aata hai. Annelida (segmented worms, true coelom) mein Leech, Nereis, Earthworm shaamil hain. *[Q1032]*

22. Reptile — Heart Structure

- Zyadatar reptiles ka '3-chambered heart' hota hai (2 atria + 1 partially divided ventricle) — Crocodile exception hai (4-chambered). Birds-Mammals 4-chambers, Fish 2-chambers. *[Q1035]*

23. Epithelium Types

- Kidney ki inner lining 'Cuboidal' epithelium se bani hoti hai — mechanical support deti hai, salivary glands ducts mein bhi hoti hai. Columnar intestine, Squamous oesophagus-mouth mein. *[Q1036]*
- Mouth ki inner lining 'Simple squamous epithelium' se bani hoti hai — constant abrasion wali areas ke liye suited. Stratified squamous skin ki outer covering banata hai. *[Q1042]*
- 'Epithelial' tissue glands banati hai — body surfaces cover karti hai, body cavities-hollow organs line karti hai, functions: protection, secretion, absorption, excretion, filtration. *[Q1044]*

24. Connective Tissue — Ligament, Tendon & Areolar

- 'Ligament' bones ko connect karta hai — bone-to-bone connection, joints-organs ko jagah par hold karta hai. *[Q1046]*
- Humari bones 'Connective' tissue hain — extracellular matrix aur limited cells se bani hoti hain (adipose, cartilage, blood bhi connective tissue hain). *[Q1047]*
- Areolar tissue organs ke andar (inside) spaces fill karti hai, internal organs ko support karti hai aur tissue repair karti hai — skin-muscles ke beech, blood vessels-nerves ke around, bone marrow mein bhi hoti hai. *[Q1048]*

- 'Tendons' muscles ko bones se connect karte hain — tough fibrous connective tissue, movement facilitate karte hain. Ligaments bone-to-bone connect karte hain. *[Q1062]*
- 'Muscle' tissue minimal extracellular matrix wali connective tissue hai — densely packed cells jo contract karne ki special ability rakhte hain (blood-bone ke muqable minimal matrix). *[Q1071]*
- Adipose (fatty) tissue skin ke neeche, kidneys ke around, aur internal organs ke beech milti hai — fat store karke insulation-protection deti hai. *[Q1072]*
- ⚠ Trap: 'Muscle' tissue insulator ki tarah kaam NAHI karti — 'Fatty' (Adipose) tissue insulator hai, energy fat ki tarah store karti hai, body temperature maintain karti hai. *[Q1074]*

25. Adipose Tissue

- 'Adipose' tissue skin ke neeche aur internal organs ke beech banti hai — loose connective tissue hai, fat store karti hai, insulation deti hai, major organs protect karti hai. *[Q1043]*

26. Aves — Birds

- 'Aves' group mein Birds shaamil hain — pelicans, ostriches, hawks, eagles, owls examples hain. Sabhi birds ke feathers, 4-chambered heart, wings, beaks hote hain, hard-shelled eggs dete hain. *[Q1045]*

27. Echinodermata — Free-Living Marine Animals

- 'Echinodermata' completely free-living marine organisms hain — water vascular system rakhte hain locomotion-food transport-respiration ke liye, koi excretory system nahi hota. *[Q1053, Q1077]*
- Starfish 'Echinodermata' group ke hain — radial symmetry, calcium carbonate ka spiny endoskeleton, water vascular system. *[Q1056]*
- 'Water vascular system' Sea cucumber (Echinodermata) mein milta hai — Sea anemone-Sea Pen phylum Cnidaria mein aate hain. *[Q1069]*

28. Porifera — Immobile Sponges

- 'Porifera' (sponges) ke animals immobile (sessile) hote hain adults ki tarah — mostly marine, asymmetrical, primitive multicellular, cellular level organisation. Examples: Sycon, Spongilla, Euspongia. *[Q1054]*

29. Planaria — Simple Eyes (Eyespots)

- 'Planaria' mein bahut simple eyes (eyespots/ocelli) hote hain — light detect karte hain lekin image form nahi karte. Freshwater flatworms hain, remarkable regenerative abilities. *[Q1055]*

30. Reptile Skin

- 'Reptile' class ki skin dry aur gland-less hoti hai — scaly, cold-blooded, creeping-burrowing terrestrial animals (snakes, lizards, turtles, crocodiles). *[Q1057]*
- Reptile ki skin 'dry-rough scales' se covered hoti hai (glandless) — water loss prevent karti hai, body protect karti hai. *[Q1058]*
- ⚠️ Trap: 'Toad' reptile class ka NAHI hai — amphibian hai (dual life, porous skin). Turtle, Crocodile, Snake true reptiles hain (dry-scaly skin). *[Q1075]*

31. Mammal Skin

- Mammals ki skin 'glandular with hairs' hoti hai — sweat, scent, sebaceous glands hote hain, hair-fur se covered hote hain, diverse environments mein adapt karte hain. *[Q1059]*

32. True Marine Fish — Dog Fish

- 'Dog fish' true marine fish hai — class Chondrichthyes (sharks wali), cartilaginous skeleton. Jellyfish-Starfish-Silverfish true fish NAHI hain. *[Q1060]*

33. Amphibia

- 'Amphibia' class ke animals scales NAHI rakhte, mucus glands hoti hain skin mein — cold-blooded vertebrates, land-water dono mein rehte hain, 3-chambered heart (2 auricles, 1 ventricle). *[Q1064]*

34. Match List — Body Tissues

- Match: Skin → Stratified squamous epithelium; Cartilage → Surface of joints; Skeletal muscles → Striated muscles; Subcutaneous layer → Adipose tissue. *[Q1066]*

35. Egg-Laying Mammals — Monotremes

- 'Platypus' egg-laying mammal hai — monotremes group mein aata hai, unique mammals jo live young ke bajaye ande dete hain. *[Q1067]*
- 'Tachyglossus' (echidna) Reptiles aur Mammals ke beech connecting link hai — dono ki characteristics rakhta hai (ande deta hai lekin fur-milk bhi hai). Peripatus annelids-arthropods connect karta hai. *[Q1068]*

36. Hydra — Reproduction by Budding

- 'Hydra' Budding se reproduce karta hai (Yeast bhi) — chhoti outgrowth (bud) se naya organism develop hota hai. Binary fission Amoeba mein, Fragmentation Spirogyra mein, Multiple fission Plasmodium mein hota hai. *[Q1076]*

37. Diploblastic vs Triploblastic Animals

- ⚠ Trap: 'Jellyfish' (Cnidaria) triploblastic NAHI hai — diploblastic hai. Triploblastic animals (3 germinal layers: ectoderm, mesoderm, endoderm) mein Earthworm, Planaria, Ascaris, Frogs, Humans aate hain. *[Q1078]*